

Table 1: Amino acid substitutions in SA14-14-2 vaccine strains, in reference to the newly-sequenced, wild-type SA14 strain.

Position	Gene	Codon	SA14 ^a	P1 ^a	P2 ^a	AF315119 ^b	D90195 ^c	JN604986 ^d
292	C	66	L	S	S	S	S	S
1296	E	107	L	F	F	F	F	F
1389		138	E	K	K	K	K	K
1503		176	I	V	V	V	V	V
1506		177	T	A	A	A	A	A
1708		244	E	G	G	G	G	G
1769		264	Q	H	H	H	H	H
1813		279	K	-	-	M	M	M
1921		315	A	V	V	V	V	V
2293		439	K	R	R	R	R	R
2317		447	G	-	-	D	-	-
3351	NS1	292	G	S	S	S	S	S
3493		339	R	M	M	M	M	M
3528		351	D	H	H	H	H	H
3539	NS2A	2	N	K	K	K	K	K
3652		40	A	V	V	V	V	V
4403	NS2B	63	E	D	D	D	D	D
4408		65	D	G	G	G	G	G
4475		87	L	-	-	F	-	-
4782	NS3	59	M	V	V	V	V	V
4921		105	A	G	G	G	G	G
5311		235	A	V	V	V	V	-
5634		343	R	-	-	-	W	-
6634	NS4A	57	I	-	-	-	T	-
7227	NS4B	106	I	V	V	V	V	V
7768	NS5	31	A	-	-	G	-	-
7809		45	R	-	-	S	-	-
8261		195	M	-	-	I	-	-
8832		386	H	Y	Y	Y	Y	Y
9593		639	Q	H	H	H	-	H
9688		671	V	A	A	A	A	A
9954		760	A	-	-	P	-	-
9978		768	L	-	-	V	-	-
10428	3'UTR	-	-	-	-	C	C	C
10574		-	-	-	-	U	-	-
10701		-	-	-	-	-	-	-
10784		-	-	-	-	-	U	-

^a Sequenced in this study.

^b Not published.

^c Reported by Aihara et. al. 1991.

^d Reported by Song et. al 2012.